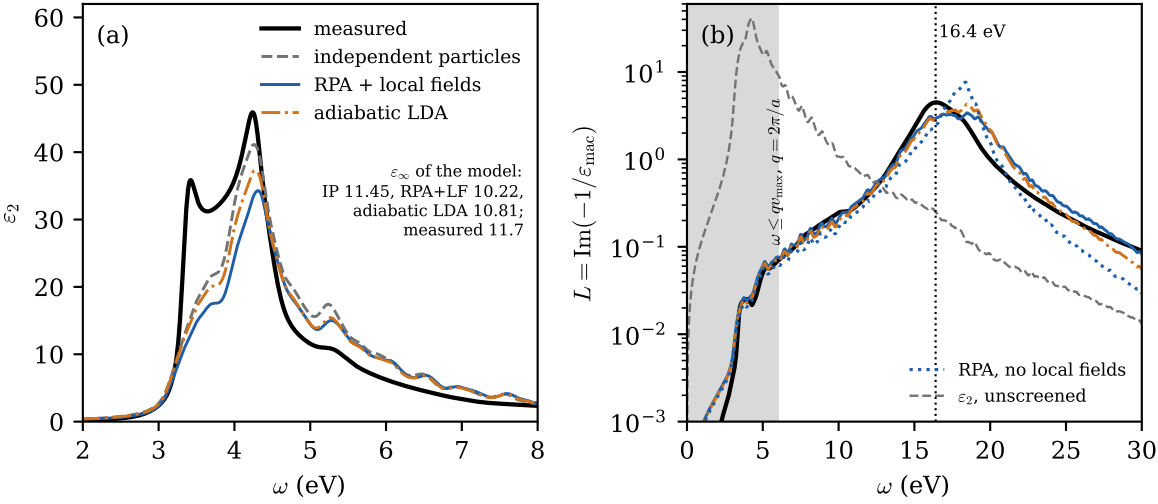
(a)
measured
independent particles
RPA + local fields
adiabatic LDA
ε_∞ of the model:
IP 11.45, RPA+LF 10.22,
adiabatic LDA 10.81;
measured 11.7
ε_2
ω (eV)
(b)
16.4 eV
$\omega \leq q v_{\max}$, $q = 2\pi/a$
RPA, no local fields
ε_2, unscreened
$L = \mathrm{Im}(-1/\varepsilon_{\mathrm{mac}})$
ω (eV)